

LaTeX to Plain Text — Demo

This file contains various LaTeX math expressions that test the full conversion pipeline.

Greek Letters

$$\alpha + \beta = \gamma$$

$$\Gamma, \Delta, \Theta, \Lambda, \Xi, \Upsilon, \Sigma, \Psi, \Phi, \Omega$$

Calculus & Analysis

$$\int_0^\infty e^{-x} dx = 1$$

$$((\partial^2 f)/(\partial x^2)) + ((\partial^2 f)/(\partial y^2)) = 0$$

$$\nabla \cdot \mathbf{F} = ((\partial F_x)/(\partial x)) + ((\partial F_y)/(\partial y))$$

Arrows and Relations

$$x \rightarrow \infty \implies (1/x) \rightarrow 0$$

$$A \subset B \subseteq C \implies A \subseteq C$$

$$f: \mathbb{R} \rightarrow \mathbb{C}$$

Fractions and Roots

$$((-b \pm \sqrt{b^2 - 4ac})/2a)$$

$$\sqrt[3]{(x^2 + y^2)} \neq \sqrt{(x^2)} + \sqrt{(y^2)}$$

Sums and Products

$$\sum_{n=1}^\infty (1/(n^2)) = ((\pi^2)/6)$$

$$\prod_{k=1}^n k = n!$$

Set Theory

$$\{x \in \mathbb{R} \mid x^2 < 2\}$$

$$A \cap B = \emptyset \implies A \text{ and } B \text{ are disjoint}$$

Trigonometry

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

$$\tan(\theta) = (\sin(\theta)/\cos(\theta))$$

Limits

$$\lim_{x \rightarrow 0} (\sin(x)/x) = 1$$

$$\lim_{n \rightarrow \infty} (1 + (1/n))^n = e$$

Matrices

$$a \ b ; c \ d$$

$$\det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = -2$$

Aligned Equations

$$E = mc^2 ; F = ma ; \nabla \times E = -(\partial B)/(\partial t)$$

Cases

$$f(x) = x^2 \text{ if } x \geq 0 ; 0 \text{ if } x < 0$$

Combinatorics

$$C(n, k) = ((n!)/(k! (n-k)!))$$

Mod

$x \equiv y \pmod{n}$

Subscripts and Superscripts

$$x_1^2 + x_2^2 = r^2$$

$$e^{i\pi} + 1 = 0$$

Code Blocks

A regular code block:

```
def fibonacci(n):  
    if n <= 1:  
        return n  
    return fibonacci(n-1) + fibonacci(n-2)
```

And some inline `code` with **bold** and *italic* formatting.

That's all folks!